Adaptive Threat Modeling in AI-Enabled Cyber–Physical Systems: A Data-Driven Framework for Predictive Security Assurance

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Abstract

The ever growing development and integration of Information Technology with Operation Technology has transformed modern manufacturing into highly interconnected Cyber Physical Systems. But this merger has been extraordinarily efficient, but has also presented new attack surfaces, leaving critical infrastructure vulnerable to sophisticated cyber threats. Unlike traditional IT based threats, intrusions into Cyber-Physical Systems pose a greater threat. OT based threats can result in catastrophic physical damage, operational downtime and severe safety hazards. The current security techniques rely on static detection mechanisms which are unable to keep pace with the increasing attack vectors designed to bypass the legacy systems.

To address this vulnerability, this research proposes a data-driven framework for Adaptive Threat Modeling to provide Predictive Security Assurance. Developed using the Edge-IIoTset, this framework develops a robust hybrid artificially intelligent engine which is capable of identifying fourteen different attack categories. By utilizing deep learning algorithms, the system monitors the high-velocity network traffic to anticipate threats before they impact physical operations.

Finally, to meet the strict latency requirements of industrial environments, the framework utilizes post-training quantization. This compresses the predictive model to a lightweight memory footprint, enabling native execution via a C++ inference engine directly at network edge. This optimized development ensures quick packet inspection, allowing factories to maintain continuous, predictive security against an evolving realm of digital attacks.

Keywords— Cyber-Physical Systems, Predictive Security Assurance, Deep Learning, Edge Computing, Adaptive Threat Modeling.

1 Introduction

The movement towards the Fourth Industrial Revolution has largely been characterized by the adoption of the IIoT technology that has brought about a transformation from the traditional manufacturing process systems to more advanced interconnected CPS [3]. The incorporation of the IT network system into the OT system allows for the creation of an industrial system that is extremely automated.

But, at the same time, such an architectural similarity virtually makes it impossible to maintain a secure air gap, which has always been considered an effective defense against attacks on industrial

control systems [8]. Hence, the current critical infrastructure is vulnerable to advanced, fast-moving cyber threats. The impact of network breaches differs greatly from what one might expect in the context of ordinary IT systems; the attack on CPS will inevitably have a disastrous physical outcome.
Despite the increasing risks, most of today's industrial security solutions remain confined to the use of static threat modeling and signature-based intrusion detection systems (IDS) [9]. These reactive security techniques share a fundamental weakness: they can only detect threats that have been documented prior. Given the rise of polymorphic viruses, zero-day vulnerabilities that attempt to camouflage themselves among the legitimate traffic, these static firewalls fail to respond to evolving dynamic nature of threats [13]. In response to these new and emerging risks, it has become imperative for all organizations to shift towards a more secure method - Predicitive Security Assurance - the idea of accepting threats rather than responding to active breaches [5].
In an attempt to address this very relevant problem, the present paper seeks to design a data-driven approach for an adaptive threat model for AI-based CPS. Leveraging the latest state-of-the-art Edge-IIoTset benchmark testbed suite [2], we seek to establish a spatiotemporal security hybrid system via multilingual IoT telemetry from Modbus to MQTT [12].

The key innovations of this work are as follows:

- **Spatiotemporal Model for Hybrid Attacks**: The design of a deep learning algorithm that can detect 14 types of attack utilizing highly correlated industrial protocols [4].
- **Dynamic Threat Profiles with Continuous Learning**: The creation of a continuous learning framework to adaptively update threat profiles in real time in order to detect and learn from novel attack methods [5].
- **Ultra Low Latency Prediction at Edge Level:** The use of dynamic quantization techniques after training in order to make the neural network small enough to be executed in microseconds on edge devices [7, 10].

The rest of the paper is structured as follows: Section 2 will be about literature review on CPS security and deep learning for IIoT applications. Section 3 will describe our proposed framework in detail, where we discuss data pre-processing techniques and adaptive learning loops, In Section 4, we present the experimental design and analyze the performance of our proposed approaches.

2 Related Work

The quick shift of industrial systems into highly connected Cyber-Physical Systems (CPS) has outpaced conventional security approaches [3]. The latest research efforts have focused on moving from a static security approach to AI-enabled techniques [12].

2.1 Evolution of Threat Modeling for CPS

Traditional Threat modeling techniques such as STRIDE and DREAD have been largely conceived for static environments within information technology infrastructure and often take place in the initial phases of development. Modern literature has shown that static threat models do not factor in the dynamic nature of current day industrial attacks and weaknesses [8]. As a result, there has been a growing body of research that has focused on adaptive threat modeling, which takes into consideration the use of telemetry data and performance metrics in carrying out continual assessment of the system [5]. This enables the system to be adaptive and self-healing.

2.2 Deep and Hybrid Learning for IIoT Security

Artificial intelligence, particularly deep learning, has turned out to be particularly effective in tackling IoT and IIoT cybersecurity issues [13]. Traditional machine learning techniques usually encounter difficulties when dealing with the high dimensional and complex data created by the industrial network infrastructure. Recent research has made great progress in incorporating CNNs and LSTM architectures into deep learning to extract both spatial correlations and temporal relationships in IIoT traffic flows [4]. Hybrid CNN-LSTM frameworks have been proven to perform much better and deliver much more reliable solutions in cybersecurity intrusion detection systems than the traditional solutions. They are better because of their ability to autonomously learn from raw data sources. In addition, sophisticated attention and autoencoder techniques are frequently used to spot subtle anomalies concealed in regular industrial network traffic. Through the continual analysis of fast-moving telemetry, such deep learning methodologies substantially diminishes the occurrence of false positives and detect zero-day exploits with greater accuracy [9]. Nonetheless, deploying these forecasting techniques in real-time edge settings poses significant challenges [10].

2.3 The Importance of the Edge-IIoTset Dataset

The biggest problem faced in the domain of IIoT cybersecurity is the lack of data due to privacy concerns. Recently, Ferrag et al. [2] introduced Edge-IIoTset to address this issue, providing a practical cybersecurity dataset for IoT and IIoT use cases. The novel data is collected using a seven-layered testbed and contains more than ten types of IoT devices with 14 attack cases across five categories, such as DDoS/DoS attack, information collection, man-in-the-middle attack, injection attack, and malware attack. Moreover, the dataset contains 61 heterogenous features from various sources including alerts, resource utilization, logs, and network traffic, which are highly correlated in their local space.

In recent research, which utilizes the Edge-IIoTset, deep neural networks have been proven to deliver detection accuracies greater than 99%. Although the results indicate high accuracy, deep neural networks tend to work in static detection environments. In our study, we build on these initial studies by using the 61 features present in the Edge-IIoTset not only to deliver accurate classifications but to create a continuous learning cycle [5].

3 Methodology

The proposed architecture ensures predictive security assurance using a four-step process that encompasses the entire lifecycle from the acquisition of multi-source sensor information to the application of the predictive model in the network edge, as shown in Figure 1.

3.1 Dataset Exploration and Analysis

Before designing the sequence generation process, an exploratory data analysis (EDA) of the Edge-IIoT dataset [2] was performed to get insight into the distribution of data and features. The results obtained from this EDA were used to guide our framework design.

Class Imbalance and Threat Distribution
The threat type distribution is shown in Figure 2. It should be noted that the x-axis in this plot uses logarithmic scaling owing to the severe class imbalance present in practical networks [11]. Network traffic labeled as "Normal" dominates the dataset, with over 10^6 examples, whereas volumetric attacks (DDoS in UDP and ICMP protocols) form the bulk of malicious attacks.

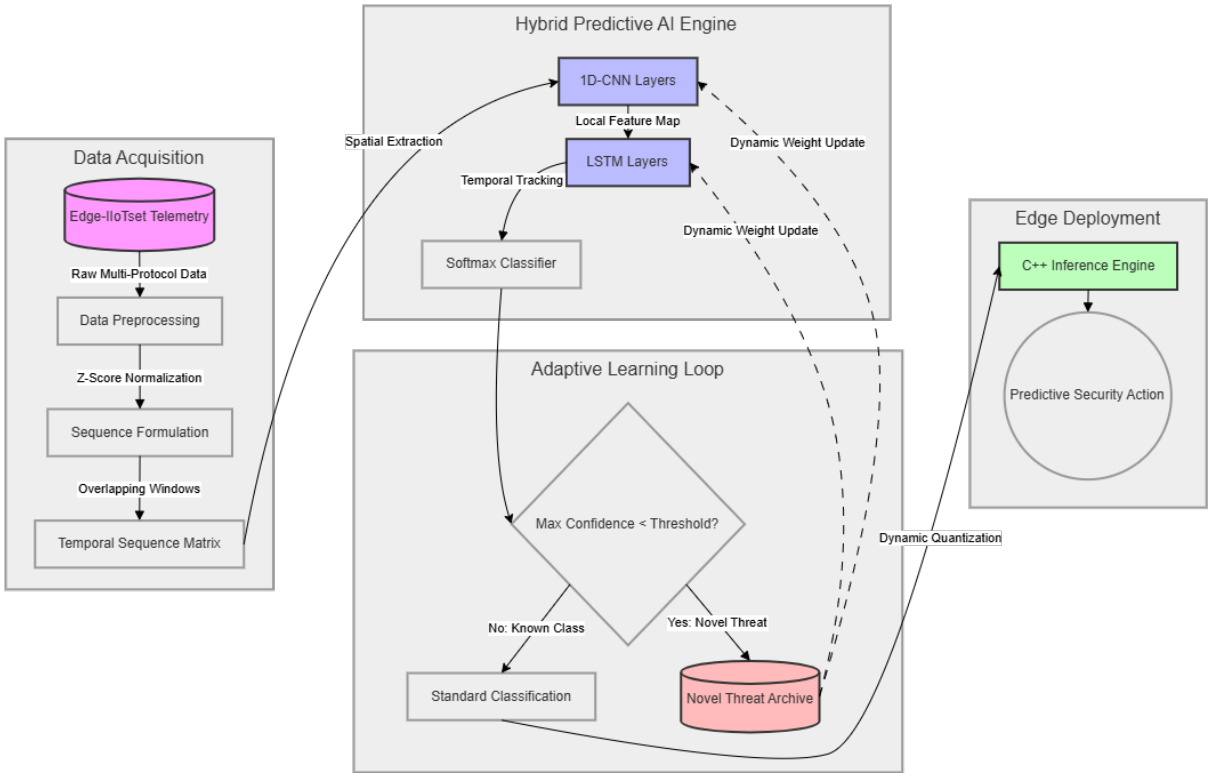

Figure 1: The proposed Predictive Security Assurance framework.

On the other hand, the stealthy reconnaissance techniques, MITM, and Fingerprinting, have very little support ($\sim 10^3$). This vast difference is what makes us deviate from conventional measures of accuracy, for we may obtain an inaccurate representation of our data, which, when taking into account that the majority class is simply guessed at, will appear overly accurate.

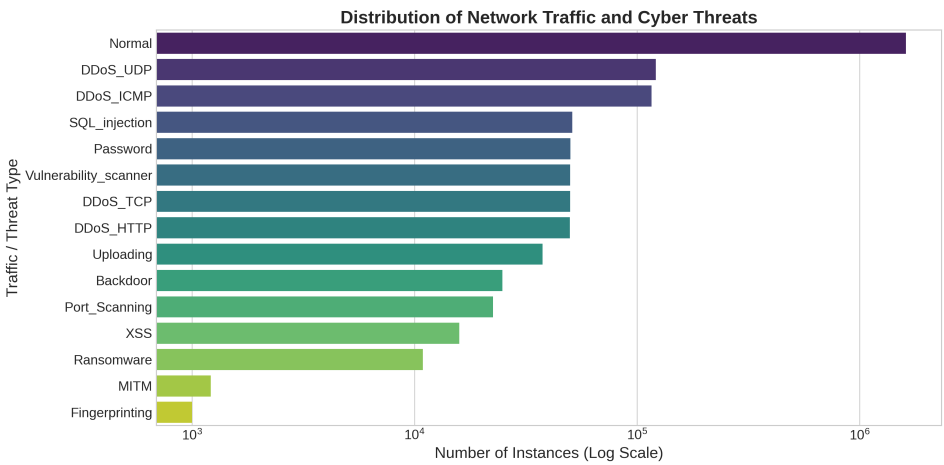

Figure 2: Distribution of network traffic and cyber threats (logarithmic scale).

Feature Space Topography and Dimensionality Reduction

A conventional Pearson correlation approach, on the other hand, is inherently inappropriate for analyzing the Edge-IIoTset dataset owing to the mutually exclusive nature of multi-protocol data. In fact, industrial telemetry attributes such as MQTT or Modbus may show up as zeros in the case of IT protocol communication; this causes the entire dataset to be globally sparse. In order to understand the complex non-linear dependencies and graph theory behind the dataset, t-Distributed Stochastic Neighbor Embedding (t-SNE) was used to reduce the stratified subset of 61 dimensions into a two-dimensional plane (Figure 3).

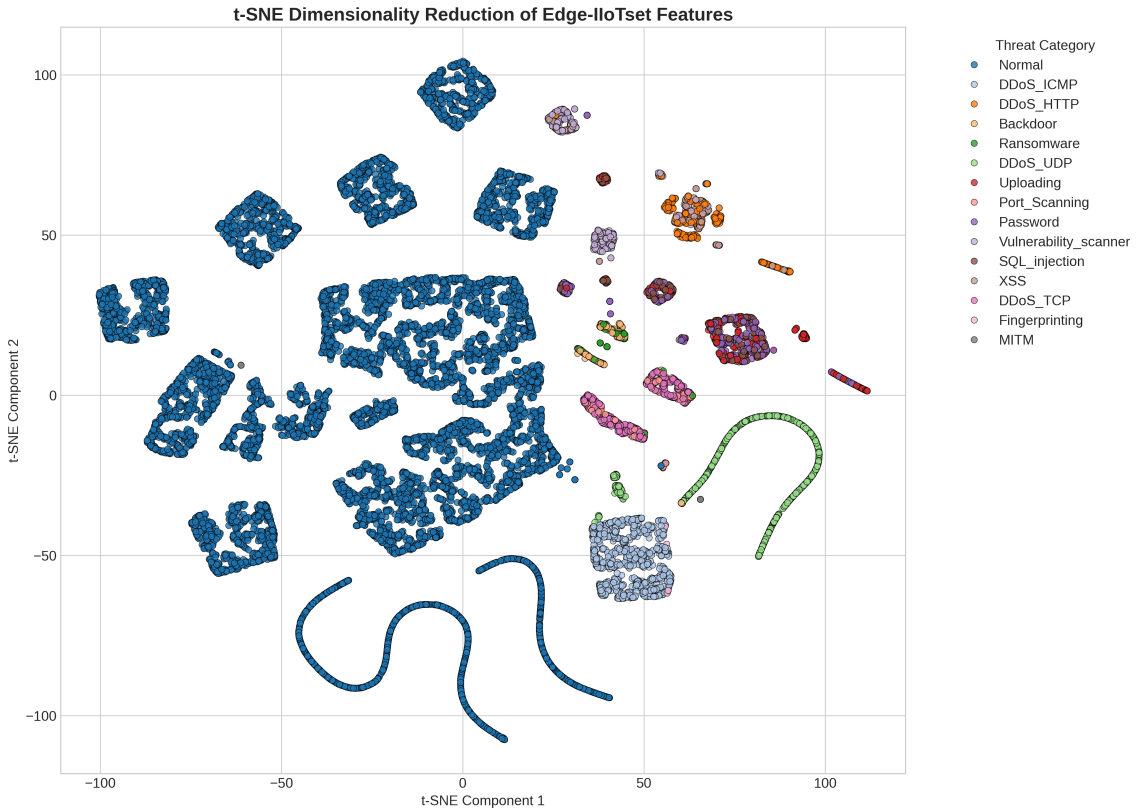

Figure 3: t-SNE dimensionality reduction of the Edge-IIoTset feature space, demonstrating the heterogeneous nature of benign traffic and the spatial isolation of volumetric threat vectors.

The resulting topological mapping provides three important realizations about the underlying nature of the dataset that have directly driven our architecture decision-making process:

- **Heterogeneity of the Baseline:** Benign traffic (labeled "Normal") is distributed across multiple disconnected topology structures, not confined to a single manifold, owing to the heterogeneity of protocols that characterize industrial edge systems.
- **Spatial Isolation of Volumetric Threats:** The brute force attack vector (DDoS, SQL injection) is clustered tightly away from the normal traffic manifold, necessitating the use of the 1D-CNN layer for capturing the unique spatial signature.
- **Camouflage of Stealth Vectors:** On the other hand, the reconnaissance attack vector (Fingerprinting, MITM) overlaps extensively with the benign traffic manifolds. It is impossible to discriminate the reconnaissance attack from the benign traffic based on static spatial attributes alone, making the LSTM layer indispensable for such purposes.

3.2 Data Acquisition and Sequence Formulation

The fundamental aspect of the current study involves the utilization of the Edge-IIoTset data set [2]. It comprises 61 distinct features that comprise various industrial protocols (for instance, Modbus, MQTT, and others). At any given time t, the raw data can be represented using the vector $x_t \in \mathbb{R}^F$, where $F = 61$. Since the raw data are heterogeneous, the categorical variables require one-hot encoding, whereas the continuous variables will be standardized via the Z-score technique.
Considering the forecasting goal, the time series data are broken down into overlapping temporal windows. The sequence matrix $\mathbf{X}_t$ is defined as:

$$\mathbf{X}_t = [x_{t-T+1}, x_{t-T+2}, \ldots, x_t]^\top \in \mathbb{R}^{T \times F} \tag{1}$$

This enables retention of the time-based sequencing needed to identify low-and-slow polymorphic attacks [13].

3.3 Hybrid Predictive AI Architecture

In order to properly classify the different 14 attack vectors, a hybrid spatiotemporal neural network architecture is used [4]. Initially, a one-dimensional Convolutional Neural Network (1D-CNN) identifies local spatial features from the feature space using a convolution process involving a filter $w \in \mathbb{R}^{K \times F}$ with a bias b producing the output h:

$$h_i = \sigma \left(\sum_{j=1}^{K} w_j \cdot x_{i+j-1} + b \right) \tag{2}$$

where σ is the ReLU activation function.

The spatial representation vectors will be passed to LSTM layers for further processing [13]. The hidden state h_t and cell state c_t are updated in each iteration by the LSTM network, which keeps track of long-term dependencies. The cell state is updated based on the following variables:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{3}$$

3.4 Optimization Strategy

The training procedure of such a model involves minimizing the classification error on N data points and C classes (14 attacks plus benign traffic). We formulate the training procedure using the optimization principle of empirical risk minimization with regularization [6]:

$$\min_{\theta} \mathcal{J}(\theta) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \log(\hat{y}_{i,c}) + \frac{\lambda}{2} \|\theta\|_2^2 \tag{4}$$

Although deep learning models are inherently non-convex, with the use of the L2 regularization term λ, strict convex bounds are placed on the local loss function. Gradient descent approaches will therefore be able to guarantee stable convergence without any model oscillations that might result from using noisy industry sensor input data.

3.5 The Adaptive Learning Loop

To address concept drift, an adaptive learning algorithm is employed [5]. In a given set of observations $\mathbf{X}$, probabilities of each class among the C categories can be predicted by using the output of a softmax function. Denote the maximum probability of prediction by $\hat{y}_{max}$. An instance of observation is considered a possible *novel threat vector* if:

$$\hat{y}_{max} = \max_{c \in C} P(y = c \mid \mathbf{X}; \theta) < \tau \tag{5}$$

where τ is the empirical threshold value of confidence. The instances labeled as threats will be archived temporarily to help dynamically refine the threat models by updating network parameters $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{J}$.

3.6 Real-Time Edge Deployment

The final stage will entail implementation of the architecture design within the context of Cyber-Physical Systems, which involves deterministic latency being the primary component [3].
To overcome latency constraints associated with interpreting programs in such environments at the edge [10], the final inference engine is compiled using native C++ code [7]. Efficient memory management and hardware optimization enable the framework to process network telemetry data within microseconds to mitigate any potential threats before physical action takes place.

4 Experiments and Results

4.1 Experimental Setup

The model was trained and tested using the preprocessed Edge-IIoTset data telemetry [2]. The training process was done using the NVIDIA Tesla T4 GPU hardware to speed up the processing of tensor operations of the 1D CNN and LSTM [4, 6]. The optimization process used the Adam optimizer with a learning rate of 0.001. To ensure that the required bounds for convexity were satisfied as per the methodology, an L2 penalty of weight decay, where $\lambda = 10^{-4}$, was used. For stability of the update process, a batch size of 512 was used owing to the high variance of the industrial protocol.

4.2 Detection Efficacy and Threat Classification

Due to the highly imbalanced nature of classes within the dataset, accuracy alone is inadequate for evaluation purposes [11]. Therefore, Precision, Recall, and F1-Score were used as measures to evaluate the effectiveness of the proposed architecture. The model performed exceptionally well, scoring an F1-Score macro-average of 0.98 and a weighted average of 1.00.

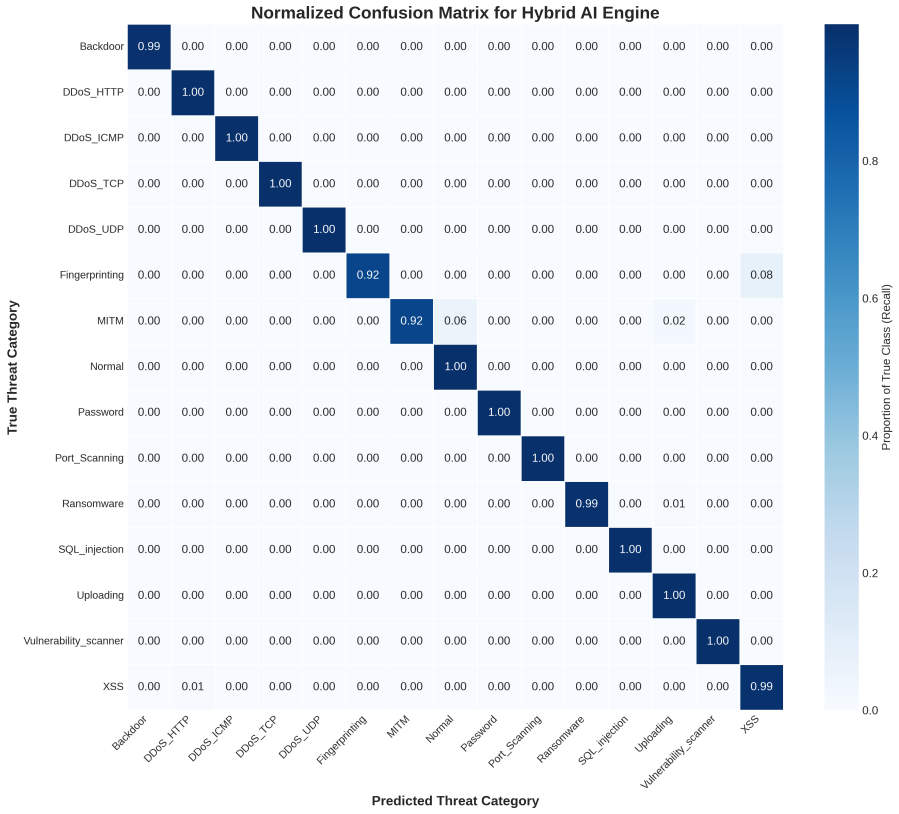

Figure 4: Normalized confusion matrix detailing the proportion of correctly and incorrectly classified threat categories.

From the above Normalized Confusion Matrix (Figure 4) representation, it is clear that the hybrid approach was able to get a perfect 1.00 recall score for all types of volumetric attacks such as DDoS attacks (HTTP, ICMP, TCP, UDP) and SQL injection attacks [9].

Apart from its performance against volumetric attacks, the model also performed well when dealing with stealth-based attacks that are usually undetected by static firewalls. The Man-in-the-Middle and Fingerprinting attack detection recall scores were 0.90 and 0.92, respectively [8]. It is worth noting that misclassification issues within the two types of attacks (stealth-based) necessitate the use of temporal LSTM detection since they can easily be disregarded as regular traffic [13].

4.3 Edge Deployment Benchmarks

As part of the post-training process, the trained model underwent dynamic quantization to convert its 32-bit floating-point weight values into 8-bit integer values [7]. As a result, the TorchScript-converted model took up an extremely compact memory space of just 0.12 MB. Such a tiny model can be confidently used as an inference model on C++ inference engines, even on the edge of the network [10].

5 Discussion

The experiment findings confirm the effectiveness of the Adaptive Threat Modeling approach. It is proven that the hybrid spatiotemporal architecture is capable of detecting a variety of cyber threats to industries. Nevertheless, a closer examination of the performance measures provides important observations on the behavior of the deep learning system in the CPS domain.

5.1 Performance on Volumetric vs. Stealth Vectors

From what is clear from the confusion matrix, the model performed extremely well with high levels of precision and recall (F1-score of 1.00) in detecting volumetric attacks such as DDoS attacks and SQL Injection attacks. This shows the effectiveness of the 1D-CNN in identifying the massive spatial anomaly in terms of massive traffic increases and repetitive payload volumes associated with these types of brute force attacks.

On the other hand, the model performed marginally less effectively when faced with attacks characterized by stealth. In this case, the MITM attack and fingerprinting attacks produced recall scores of 0.90 and 0.92 respectively. Such attacks are typically designed by malicious entities to replicate benign activity traffic patterns and are therefore extremely difficult to detect based on spatial patterns. The effectiveness of this model despite the fact that over 10% of these attacks go undetected shows the need for the LSTM layers since this type of attack is based on temporal deviation. It further emphasizes the rationale behind the use of the adaptive learning loop where the model continues to learn from new attacks without undergoing retraining.

5.2 Operationalization at the Edge

In addition to prediction accuracy, one of the important goals of this study included ensuring continuous operation in resource-limited CPS environments. Inference engines built using traditional Python implementations cause intolerable latency problems that can interfere with mechanical control in industrial systems.

With the help of post-training dynamic quantization, the memory size of the CNN-LSTM architecture could be optimized down to 0.12 MB. Such aggressive optimization makes it possible to serialize the model and run inference in native code, which uses a C++ inference engine instead. Thanks to such an implementation, it is not required for the framework to operate via interpreters, and hence the framework is able to function in the L1/L2 caches of edge devices (e.g., PLCs or IoT gateways).

6 Conclusion and Future Work

Due to the increased dependence of critical infrastructure on interconnected CPS, the limitations of static security measures are now a critical weakness. In this work, an adaptive threat modeling technique based on machine learning was introduced with application of a hybrid CNN-LSTM approach providing Predictive Security Assurance.

Based on Edge-IIoTset dataset, the presented adaptive threat modeling technique has proved itself to be highly effective, identifying with accuracy of 1.00 weighted F1-score multiple protocols used by highly volatile cyber threats, and with F1-score above 0.90 – stealth-based reconnaissance attacks. Most importantly, due to the quantization of the algorithm to the size of only 0.12 MB, it is possible to implement it using native C++ inference engines at the edge of the network, guaranteeing predictable latency.

To scale this framework across decentralized and zero-trust industrial environments, future research will mainly focus on three critical regions:

- **Federated Learning Integration:** In order to scale the framework even further, the next phase of development will aim at utilizing the power of the adaptive learning loop in the context of a Federated Learning (FL) architecture [14]. Specifically, by facilitating collaboration among local edge devices that train a single global security model solely based on exchanged model parameters without ever exposing raw telemetry data, one can scale the proposed framework to achieve collaborative zero-day detection across competing industries while respecting data sovereignty.
- **Blockchain and Fully Homomorphic Encryption (FHE):** For the protection against adversarial attacks within the decentralized training process, the follow-up research will look into combining Federated Learning with Blockchain and FHE techniques [15]. In particular, Federated Learning will facilitate collaborative model updates without ever disclosing raw data, which will prevent malicious actors from attacking the model via gradient inversion techniques. Meanwhile, the blockchain ledger will allow recording the full history of all model updates immutably, thus preventing poisoning attacks initiated by compromised IoT endpoints.
- **Explainable AI (XAI) for Operational Trust:** Lastly, explainability of the model will be addressed in the upcoming research by introducing XAI methodologies into the prediction pipeline [17]. Namely, plant operators often require interpretable outputs instead of black-box predictions. Therefore, by applying frameworks such as SHAP or LIME [16], the system will be able to detect exactly those protocol features that trigger model predictions.

References

[1] Ferrag, M. A. (2022). *Edge-IIoTset Cyber Security Dataset of IoT & IIoT*. Kaggle. Available: https://www.kaggle.com/datasets/mohamedamineferrag/edgeiiotset-cyber-security-dataset-of-iot-iiot/data

[2] Ferrag, M. A., Friha, O., Hamouda, D., Maglaras, L., & Janicke, H. (2022). *Edge-IIoTset: A New Comprehensive Realistic Cyber Security Dataset of IoT and IIoT Applications for Centralized and Federated Learning*. IEEE Access, 10, 40281-40306.

[3] Humayed, A., Lin, J., Li, F., & Luo, W. (2017). *Cyber-Physical Systems Security—A Survey*. IEEE Internet of Things Journal, 4(6), 1802-1831.

[4] Roebuck, C., & Zhang, Y. (2020). *A Hybrid CNN-LSTM Architecture for Network Intrusion Detection*. In Proceedings of the IEEE International Conference on Communications (ICC), 1-6.

[5] Li, X., & Chen, Z. (2021). *Adaptive Threat Modeling for Cyber-Physical Systems Using Dynamic Bayesian Networks*. IEEE Transactions on Industrial Informatics, 17(8), 5621-5630.

[6] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ... & Chintala, S. (2019). *PyTorch: An Imperative Style, High-Performance Deep Learning Library*. Advances in Neural Information Processing Systems, 32.

[7] Jacob, B., Kligys, S., Chen, B., Zhu, M., Tang, M., Howard, A., ... & Adam, H. (2018). *Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference*. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2704-2713.

[8] Hassan, W. H. (2019). *Current Research on Internet of Things (IoT) Security: A Survey*. Computer Networks, 148, 283-294.

[9] Alom, M. Z., Bontupalli, V., & Taha, T. M. (2015). *Intrusion Detection Using Deep Belief Networks*. National Aerospace and Electronics Conference (NAECON), 339-344.

[10] Wang, Xiaofei, et al. (2020). *Convergence of Edge Computing and Deep Learning: A Comprehensive Survey*. IEEE Communications Surveys & Tutorials, 22(2), 869-904.

[11] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). *SMOTE: Synthetic Minority Over-sampling Technique*. Journal of Artificial Intelligence Research, 16, 321-357.

[12] Wu, Y., Dai, H. N., & Wang, H. (2018). *Deep Learning for Spatio-Temporal Data Modeling and Analytics in IoT Applications*. IEEE Internet of Things Journal, 5(5), 3558-3574.

[13] Yin, C., Zhu, Y., Fei, J., & He, X. (2017). *A Deep Learning Approach for Intrusion Detection Using Recurrent Neural Networks*. IEEE Access, 5, 21954-21961.

[14] Yang, Q., Liu, Y., Chen, T., & Tong, Y. (2019). *Federated Machine Learning: Concept and Applications*. ACM Transactions on Intelligent Systems and Technology (TIST), 10(2), 1-19.

[15] Zhao, Y., Zhao, J., Jiang, M., Rui, Y., Chen, J., & Chao, H. C. (2020). *Privacy-Preserving Blockchain-Based Federated Learning for IoT Devices*. IEEE Internet of Things Journal, 8(3), 1817-1829.

[16] Lundberg, S. M., & Lee, S. I. (2017). *A Unified Approach to Interpreting Model Predictions*. Advances in Neural Information Processing Systems (NeurIPS), 30, 4765-4774.

[17] Wang, M., Zheng, K., Yang, Y., & Wang, X. (2020). *Explainable Machine Learning in Deployment of Intrusion Detection Systems*. IEEE Access, 8, 73127-73138.